TinyStar'Ex

Spectrograph build and assembly guide

Version 1.1

A STAROS project

staros-projects.org

1 Purpose

This document is the build, assembly and adjustment guide for the `TinyStar'Ex` (TSE) spectrograph. Figure 1 shows the finished instrument and gives an idea of its scale (the pointing system and the cameras are not fitted).

Figure 1. The TinyStar'Ex (TSE) spectrograph unit held in the hand.

The front of the spectrograph is a tube 40 mm in diameter (39.75 mm, to be precise) that carries the entrance slit. This tube also serves as the 'universal' interface of the TSE system, onto which accessories can be fitted (Figure 2).

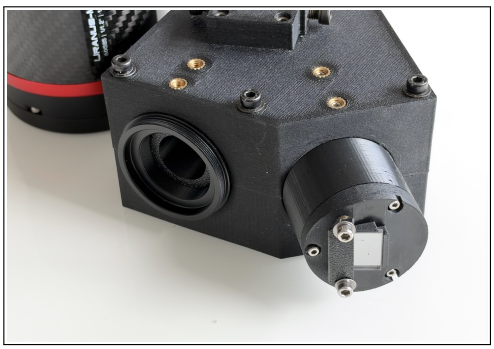

Figure 2. The front of the spectrograph carries a four-position slit, offering a choice of four widths. The rear is a standard male T2 interface that accepts any type of camera.

One of these accessories is essential for a fully operational instrument on the sky: the system for pointing at the target and guiding on the slit. Its construction is described in a separate document.

The rear of the instrument is a male T2 threaded interface. Any camera with a back focal distance of 17.5 mm, a standard value, can be fitted there and used directly to record spectra.

2 Licence

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The `TinyStar'Ex` design files (STL print files, OpenSCAD sources and instruction manuals) are licensed under the *Creative Commons Attribution–NonCommercial–ShareAlike 4.0 International licence (CC BY-NC-SA 4.0)*. The optical kit, which is sold commercially, is not covered by this licence.

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3 Internal layout of the spectrograph

TinyStar'Ex uses a so-called 'classical' optical design, the same as that of the Star'Ex LR and Star'Ex HR spectrographs, but in a more compact form.

In TinyStar'Ex, the collimator lens and the camera lens are identical. They are doublets made of special glasses that give them high optical performance in terms of achromatism, image sharpness and field coverage. These lenses have a focal length of 50 mm and a diameter of 20 mm.

The nominal diffraction grating is a 300 lines/mm model, blazed in the visible, measuring 12.7 × 12.7 mm and 6 mm thick[1].

Figure 3 details the optical layout and the dimensions of the elements.

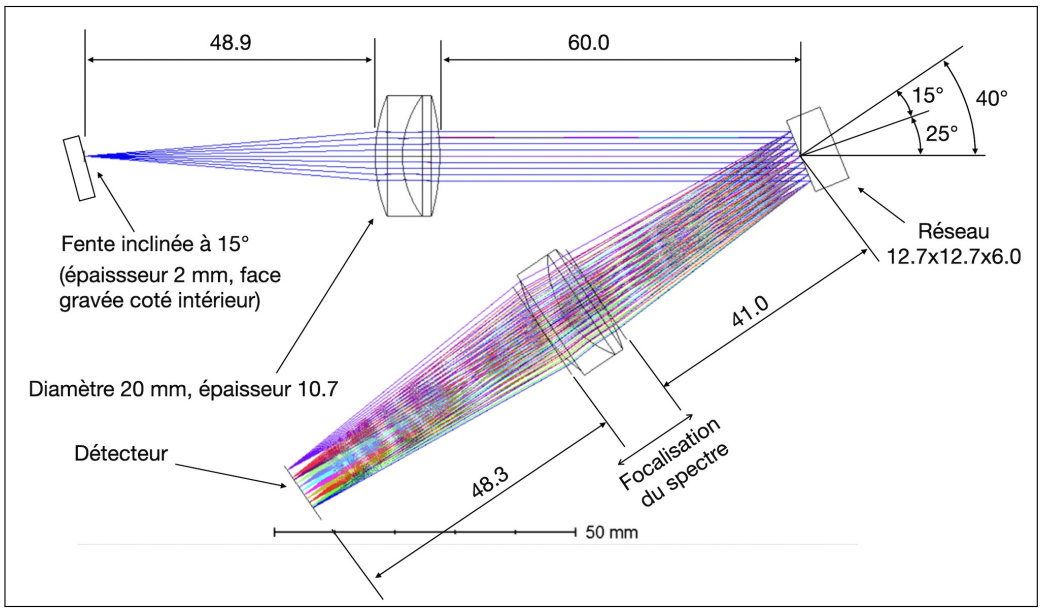


Figure 3. Optical diagram of the TinyStar'Ex (TSE) spectrograph.

Figure 4 brings together in a single image the optical components required by the spectrograph and its pointing and guiding system. Note how small these components are.

[1]The mechanical design of TinyStar'Ex allows the angle of incidence on the grating to be adjusted over a limited range, so that the spectrum can be centred precisely in wavelength on the detector. It is also entirely possible to change the grating, for example to switch to a very low resolution mode with a 150 lines/mm grating, known as SED mode ('Spectral Energy Distribution'). Spectral resolution can also be increased, over a narrower spectral range, by adopting a ruled grating of 600 lines/mm, or even 1200 lines/mm. A grating 25 mm square can even replace the standard 12.7 mm grating. All these options involving a change of grating, however, require specific holders to be printed, and their use is not covered in this guide.

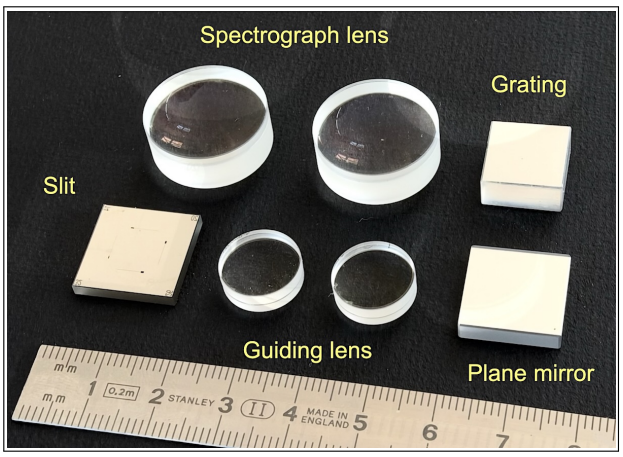


Figure 4. The optical components of the TinyStar'Ex spectrograph and its pointing and guiding system.

The slit is a four-position GEN2 model, offering a choice of widths of 14, 20, 26 and 32 μm. Each of these slits has a photometric zone at each end, where the slit is locally widened by a factor of 10.

Table 1 gives the resolving power for an extended source as a function of the slit width chosen. For a point source, the resolving power is typically 25% higher.

Table 1. Spectral resolving power for an extended source as a function of slit width, at λ = 656.3 nm (Hα line) and for an f/5 telescope.

Slit width (μm)	**Resolving power** R
14	780
20	540
26	420
32	340

The optical assembly just described is housed in a 3D-printed mechanical body. The optical layout can be recognised in the shape of this housing (Figure 5).

All the STL files needed for printing are supplied (see the next section).

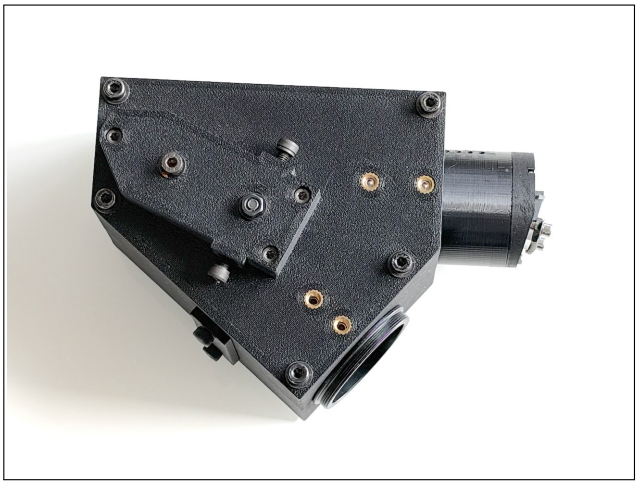


Figure 5. The TinyStar'Ex spectrograph, whose shape follows that of the optical diagram. The removable grating interface part can also be seen on top.

4 The STL files

All the STL files needed to print the spectrograph parts are gathered in a compressed archive, `stl_spectro.zip`, which can be downloaded directly from the project's GitHub repository.

Figure 6 shows all the parts to be printed to build the 'spectrograph' part of the `TinyStar'Ex` project.

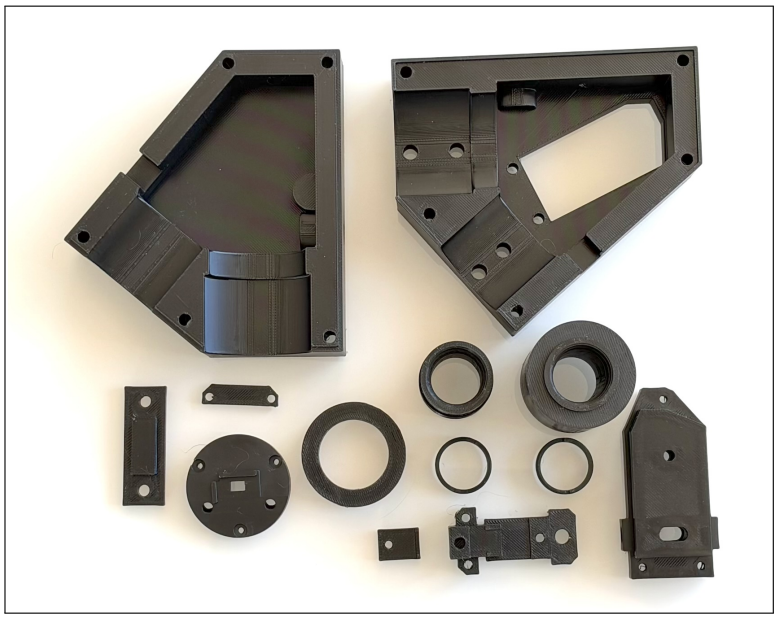


Figure 6. All the printed parts that make up the `TinyStar'Ex` spectrograph.

Section 8 gives recommendations for printing.

Table 2 is the bill of materials for the parts of the spectrograph. The part names are those of the STL files.

Table 2. Bill of materials for the printed parts of the spectrograph.

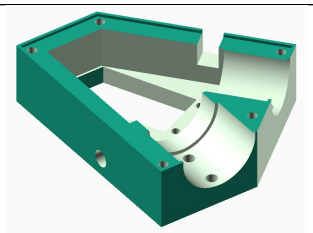
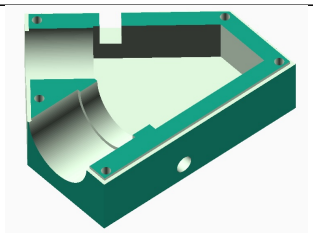
No.	Qty	View of the part
1-1-capot-inferieur	1	
2-1-capot-superieur	1	

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Table 2 – continued from previous page

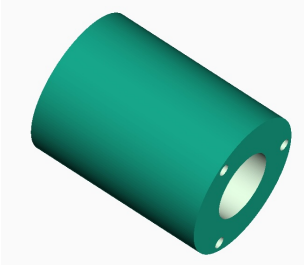
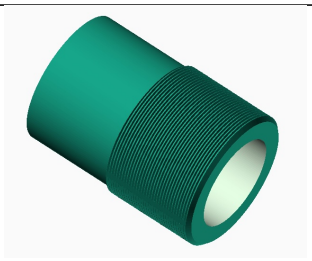
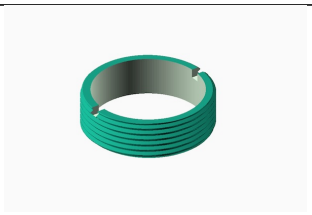
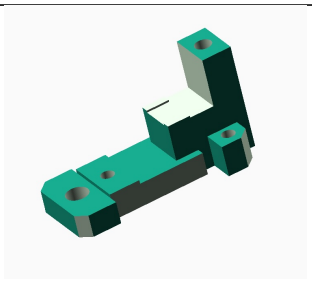
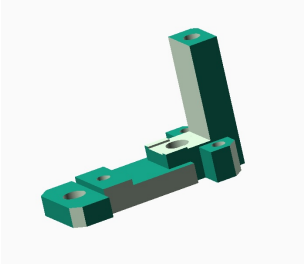
No.	Qty	View of the part
3-1-tube-collimateur	1	
4-1-tube-camera	1	
5-1-bague-objectifs	2	
6-1-bague-focus	1	
7-1-porte-reseau-127	1	
8-1-porte-reseau-250	1	

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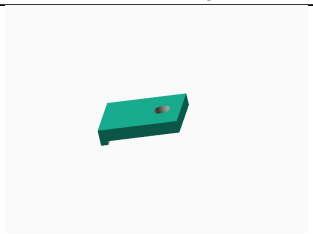
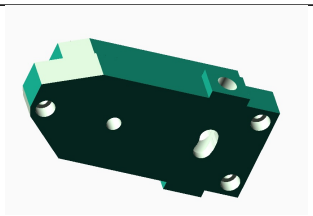
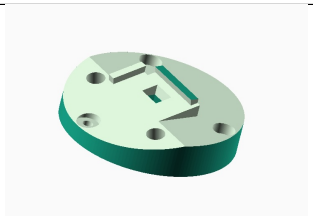
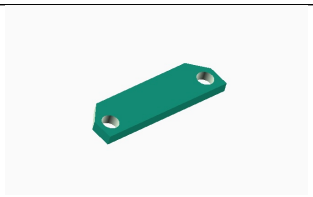
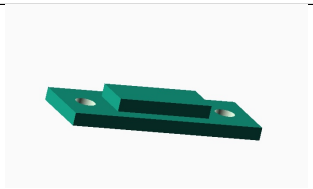
Table 2 – continued from previous page

No.	Qty	View of the part
9-1-bride-reseau	1	
10-1-interface-reseau	1	
11-1-support-fente	1	
12-1-bride-fente	1	
13-1-trappe-focus	1	

5 Adapting the parts with OpenSCAD

The STL files supplied are intended for printing in PETG. Using carbon-fibre-filled PETG (PETG-CF), or simply the particular characteristics of your printer, may require some dimensions to be adjusted slightly so that the parts fit as well as possible. This is why the project also provides an OpenSCAD file describing all the parts of `TinyStar'Ex`.

OpenSCAD[2] is free, open-source 3D modelling software, available for Windows, macOS and Linux. Unlike conventional CAD software, you do not draw with the mouse: each part is described by a short text, in the manner of a computer program. This text combines simple shapes, called primitives (cube, cylinder, sphere…), using a few operations: they are added together (`union`), subtracted from one another (`difference`) to drill a hole or cut out a recess, and moved or rotated (`translate`, `rotate`).

[2] https://openscad.org

For example, the 50 mm barrel is simply a cylinder from which a smaller cylinder is removed to let the light through. Its outer diameter is defined by a variable placed at the top of the code:

```
diametre_coulant = 50.85;   // outer diameter of the barrel (mm)
```

If the barrel is too tight a fit in your focuser, simply replace 50.85 with 50.75, for example.

In the `TinyStar'Ex` files, the important dimensions are grouped in this way into clearly identified and commented variables. The procedure is always the same:

1. open the file in OpenSCAD and change the desired value;
2. press `F5` to see the modified part immediately;
3. press `F6` to render the final part, then export it in STL format (menu *File* > *Export* > *Export as STL*).

The new STL file is ready to be sent to the slicer. There is no complex interface to learn: a few minutes are enough to adjust a diameter or a thickness. The more curious can also go further and modify, or even create, their own parts.

The OpenSCAD file containing the complete CAD model of the spectrograph, `TSE_sources_spectro.scad`, can be downloaded directly from the project's GitHub repository.

The form of `TinyStar'Ex` currently offered is fully functional, which is what matters most. You are nevertheless free to modify it to your taste and adopt your own *design language*, in the spirit of sharing the results of this work with the community. You must, of course, respect the optical dimensions (Figure 3) for the modified instrument to remain operational.

6 Screws and fasteners

One supplier (among others) for the screws: FIXnVIS, `https://www.visseriefixations.fr`.

Table 3 lists all the screws required, both for the spectrograph itself and for the pointing and guiding system, which is described in a separate guide. A word of advice: buy a few more than the quantities indicated; this is not the largest item of expenditure.

Table 3. List of screws.

Type	Quantity
M2 × 8 mm	3
M3 × 8 mm	4
M3 × 10 mm	9
M3 × 12 mm	3
M4 × 8 mm	2
M4 × 20 mm	5
M4 × 25 mm	1
M4 × 30 mm	1
M4 × 60 mm	5
M4 × 70 mm	4
M3 × 6 (grub screw)	1
M4 × 10 (grub screw)	4

Table 4 gives the type and number of threaded inserts required. Choose RUTHEX inserts. They can be bought, for example, from `https://www.3djake.fr/ruthex` or from Amazon.

Table 4. List of inserts.

Type	Quantity
M2	3
M3	12
M3 S	6
M4	14
M4 S	9
1/4" S	4

7 Tools

Only a few tools are needed (small screwdrivers, Allen keys, pliers), plus a basic soldering iron and the equipment for fitting *inserts*, the small threaded parts that allow strong screwed joints to be made (Figure 7).

Figure 7. What you need to fit inserts: a small soldering iron, special tips and a set of inserts (we recommend the Ruthex brand).

You should get a set of soldering iron tips specifically designed for installing Ruthex threaded inserts. This inexpensive accessory is essential for precise installation. For example: https://www.amazon.fr/dp/B0DP51DDF7?th=1.

The type of insert to use will be indicated at each stage of the assembly.

Set the soldering iron to a temperature of 230 °C, ideal for PETG (Figure 8).

Figure 8. Setting the temperature of the iron.

The insert should sink in gradually under light pressure (Figure 9). Do not go beyond the depth at which it is flush with the surface, to avoid raised edges. You can even stop just short of the final depth and, immediately after removing the iron, press with a metal block to push the insert flush and obtain a perfectly flat surface. The smaller the insert, the more care it needs.

Figure 9. Installing a threaded insert using a soldering iron. Apply gentle pressure without excessive force to maintain control over the insertion process. Make sure the insert is oriented correctly: insert the smaller-diameter end into the hole. If necessary, use your fingers to gently guide this end into the hole, ensuring that the insert and the hole are coaxially aligned. Only then, slowly press the insert into place using a soldering iron fitted with the appropriate tip for the insert size (M2, M3, M4, etc.), until it is flush with the surface. Apply pressure strictly along the axis of the insert.

8 Printing

Virtually any 3D printer can make a TSE, including small models such as the excellent Bambu Lab A1 mini (see Figure 10). Black PETG filament is the right choice for printing TSE: the black colour is essential for the parts to be opaque to light. If your printer allows it, you can also opt for PETG-CF (carbon-fibre filled), which offers even better rigidity.

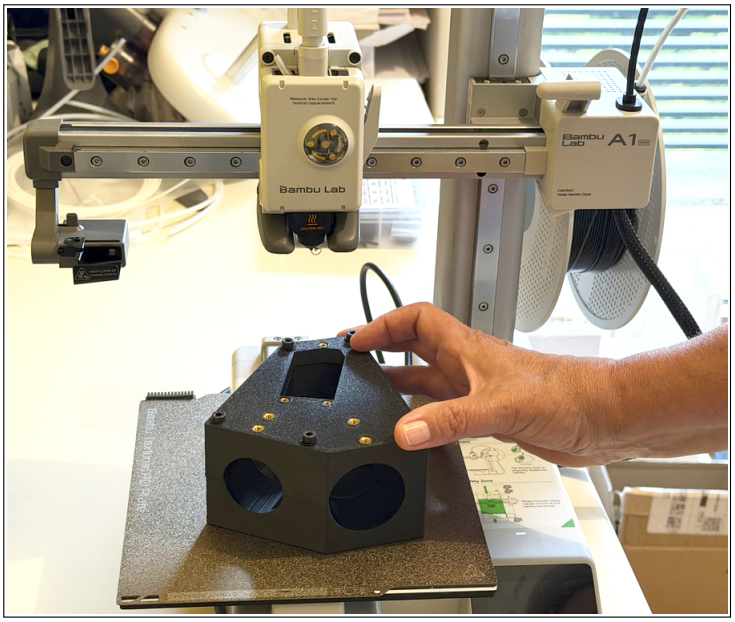


Figure 10. Despite a build plate of only 180 × 180 mm, the Bambu Lab A1 mini printer prints all the TSE parts without difficulty. Compact and now sold for less than 200 euros, such a printer is an investment that quickly pays for itself, in keeping with the philosophy of TinyStar'Ex.

Most parts can be printed with a layer height of 0.18 mm. There are, however, some important exceptions: the following parts **must** be printed with a layer height of **0.12 mm** or less (to be set in your slicer). They are identified by their number in Table 2:

- `3-1-tube-collimateur`;
- `4-1-tube-camera`;
- `5-1-bague-objectifs`;
- `11-1-support-fente`: for this part, also enable fine smoothing of the vertical walls if your printer allows it.

Five perimeters are enough. You can go up to six for the `3-1-tube-collimateur` part, which must be particularly rigid. An infill of 30% is entirely satisfactory; use a gyroid infill pattern if possible.

9 Assembling the spectrograph

9.1 The housing

Figure 11 shows all the inserts to be fitted in the lower shell (part `1-1-capot-inferieur`).

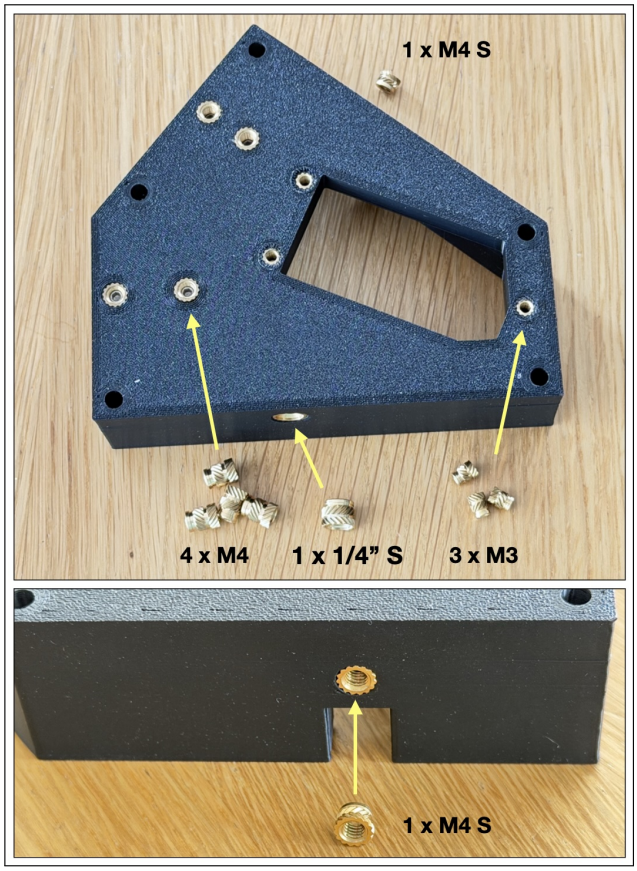


Figure 11. List and layout of the inserts to be fitted in the lower shell of the TSE housing. Take great care that the four M4 inserts do not protrude into the circular half-bore, which must remain perfectly smooth: check carefully that each insert is pushed in only until it is flush with the flat face of the housing.

The M4 inserts take Teflon-tipped grub screws (M4, 10 mm long); see Figure 12.

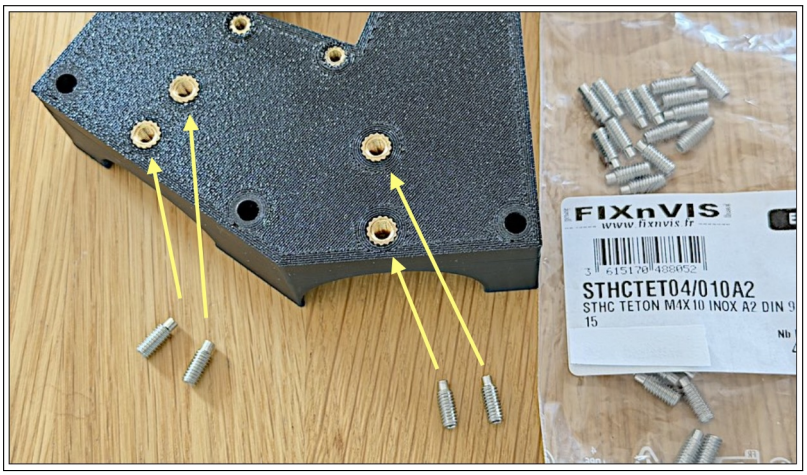


Figure 12. Locking screws for the collimator and camera tubes: Teflon-tipped grub screws (M4, 10 mm). See, for example, `https://www.fixnvis.fr`.

Figure 13 shows the inserts fitted in the upper shell (part `2-1-capot-superieur`).

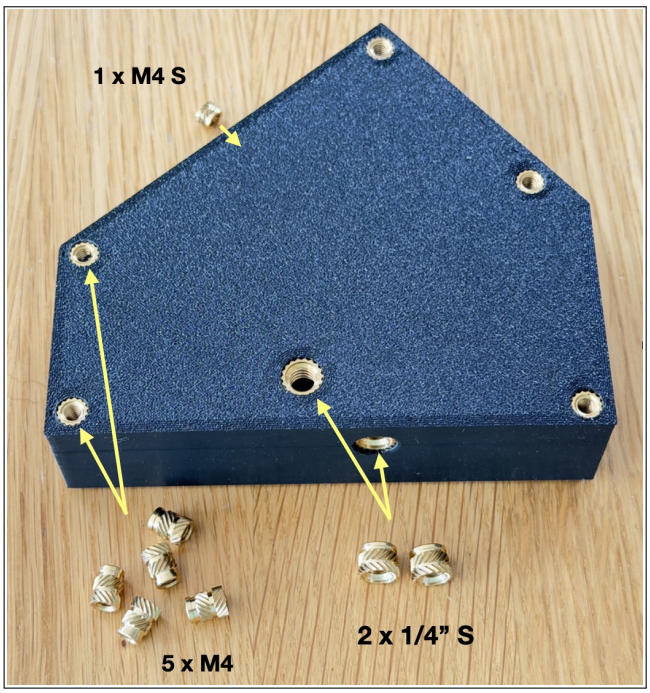


Figure 13. The inserts in the upper shell of the TSE housing. The 1/4-inch inserts (standard tripod thread) are provided so that, for example, the spectrograph can be mounted on a photographic tripod, or accessories attached to its body.

9.2 The camera tube

The camera tube holds the so-called 'camera' lens, which focuses the spectrum onto the detector of the imaging camera. To complete the assembly, you need a 31.75 mm barrel to T2 thread adapter (42 mm diameter, 0.75 mm pitch). This is a common item among astronomy accessories; you will probably find one, for example, in the box of the camera you use.

We shall use the internal thread with which all these adapters are fitted for mounting filters (28.5 mm diameter, 0.6 mm pitch). The camera tube (part `4-1-tube-camera`) screws into this adapter, which will later allow the spectrum to be focused.

Figure 14 shows how to fit these two parts together. Take care: the thread is very fine, so patiently find the start of the thread, and above all avoid forcing the tube if it is crooked in the female metal thread. When everything is correct, the tube screws in without great effort and without any tight spots. It is normal for the working clearance to be fairly small, so you may feel slight resistance.

Figure 14. Fitting the camera tube into the 1.25-inch barrel to T2 adapter. You should be able to screw it in until the threaded part of the tube disappears.

The next step is to prepare the ring that fits on the end of the camera tube, the part referenced `6-1-bague-focus`. As its name suggests, it is this ring, fixed to the camera tube, that allows the tube to be turned (through a hatch in the housing) and hence the focus to be adjusted.

A short M3 insert is fitted in the ring, so that it can be clamped with a small M3 hexagon socket screw 6 mm long (Figure 15). A Teflon-tipped screw, which does not damage the PETG, is recommended. In general, for screws and fasteners, `https://www.visseriefixations.fr` offers a very wide choice. For grub screws, type STHC in the site's search tool (see also the reference in Figure 15).

Figure 15. Fitting the clamping insert (an M3 insert) and an M3 grub screw 6 mm long in part `6-1-bague-focus`.

Fix the focusing ring almost at the end of the camera tube, as shown in Figure 16.

Figure 16. The focusing ring fixed very close to the end of the lens tube (allow about 0.5 mm between the end of the tube and the front face of the ring). Tighten the grub screw moderately.

We shall now fit the lens in its tube.

Unpack one of the two lenses from the optical kit (see the left-hand view in Figure 17). You cannot go wrong: these lenses are identical! One will serve as the camera lens and the other, later, as the collimator lens. They are 20 mm in diameter with a focal length of 50 mm.

These lenses are specially optimised achromatic doublets, made of special (ED) glasses. The two cemented elements can easily be made out by looking at the edge of the lenses.

One of the two lenses from the optical kit is fitted simply by sliding it into the camera tube (see the right-hand view in Figure 17).

Figure 17. Left: the two lenses of the optical kit, still in their packaging. Use a craft knife to cut the retaining tape when unpacking them. Right: the lens ready to be slid into its seat in the camera tube. Always hold the lens by its edge. Naturally, do not touch the optical surfaces; but if this happens, there is no cause for concern: rub gently with cotton wool and 95% household alcohol, changing the cotton wool, and everything will be fine again (99% isopropyl alcohol is even better, but not essential). You can wear white cotton gloves if you are not comfortable handling this type of component with bare hands.

Caution: the lens must be fitted the right way round, and it is **very important** to respect this. Look carefully at the enlarged view in Figure 18: the element that appears thinner when viewed edge-on must **always** face the outside of the tube.

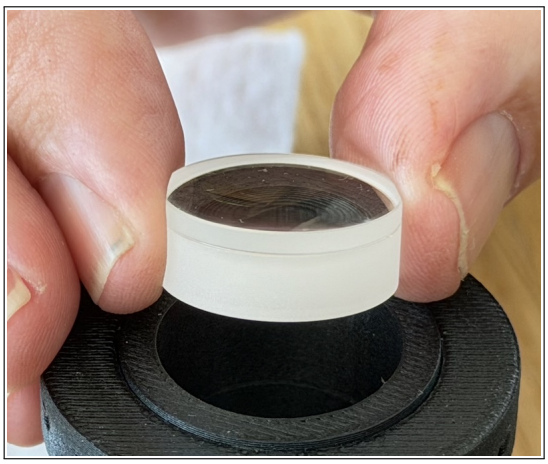


Figure 18. The orientation of the lens. It consists of two cemented elements: the thinner of the two, seen edge-on, must be placed on the outer side of the tube.

With the tube held vertically, present the lens squarely on the axis: it should drop of its own accord down to its internal shoulder, under gravity alone, without your having to press on it, except gently on one edge to straighten it. Tap the tube to help it down. If the lens has to be forced in, do not insist, as you risk jamming it in its seat. If this happens, press gently on the other face of the lens with a cotton bud to push it back out, with successive light touches on the edges and in the centre.

If you need to remove the lens from its seat, do so over lens tissue or a sheet of cotton wool, so that it cannot be damaged by falling onto a hard surface. Make sure the drop height does not exceed two or three centimetres.

All that remains is to lock the lens in its seat with one of the threaded rings `5-1-bague-objectifs` (Figure 19).

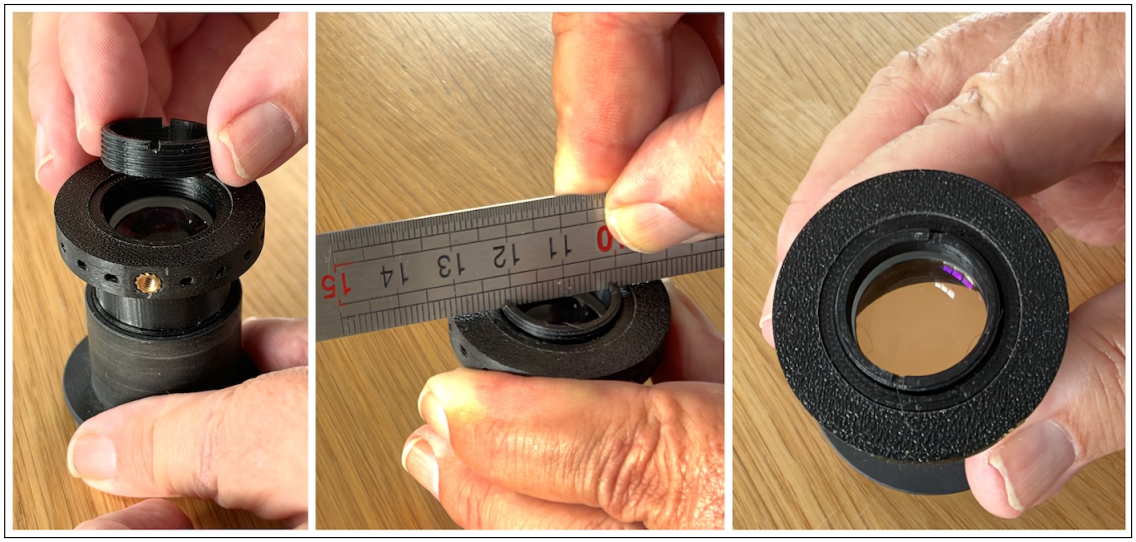


Figure 19. Screwing in the lens retaining ring. When the ring touches the glass surface, tighten it moderately by turning it with a steel rule engaged in its two notches. Once the ring is tight, the lens must no longer move when you shake the tube.

9.3 The collimator tube

Fit three M2 inserts in the seats provided at the end of the collimator tube, part `3-1-tube-collimateur` (Figure 20). This is probably the most delicate insert fitting, as they are small in diameter. Use

the appropriate soldering iron tip and push the inserts in very gently while hot. You can finish by pressing with a metal plate before they cool. The inserts must not protrude from the flat face of the collimator tube, and no raised edge must remain: the slit holder will rest on this face, and the distance between the slit and the lens, as specified by the optical design, depends on it.

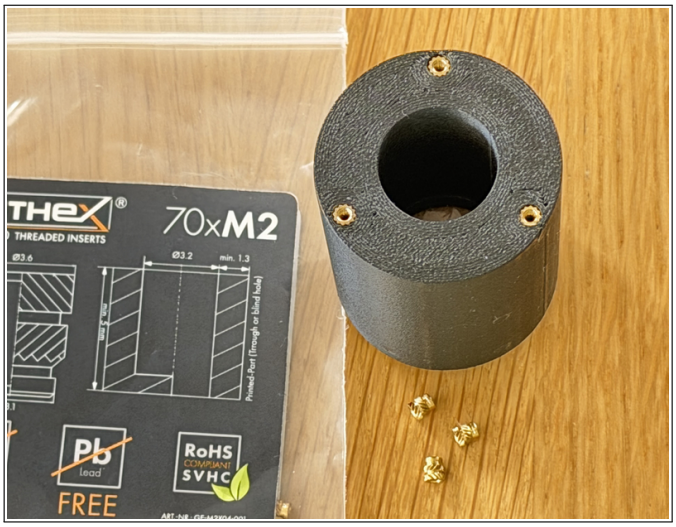


Figure 20. Fitting the M2 inserts.

Fit the remaining lens in the collimator tube in the same way as for the camera tube (Figure 21). Take particular care to fit the lens the same way round. Finally, lock the lens with a `5-1-bague-objectifs` threaded ring.

Figure 21. Fitting the collimator lens the right way round, then holding it in place with a threaded ring. As for the camera lens, finish tightening without forcing, using an engineer's rule engaged in the notches of the ring.

9.4 The slit

Take the slit holder, part `11-1-support-fente`, and fit two M3 S inserts in it (Figure 22). Take care: these are 'short' M3 inserts, not to be confused with 'standard'-length M3 inserts.

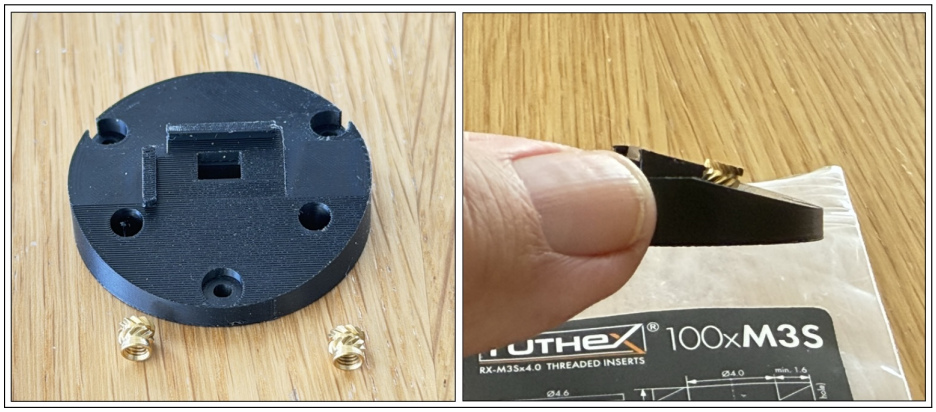


Figure 22. The slit holder and the two M3 S inserts to be fitted. Note that the plane of the slit is inclined at 15° to the base of the part; it is normal for the insert, in its fitting position, to have this angle (see the photograph on the right). When you push it in with the soldering iron, you must respect this angle, which means that the axis of the iron is perpendicular to the seating face of the slit.

Figure 23 shows the inserts in place at the correct angle.

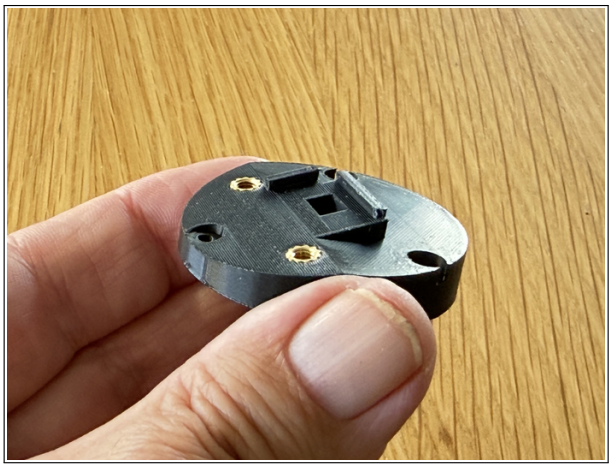


Figure 23. The slit holder fitted with its inserts. Their axis is indeed perpendicular to the flat face that carries the slit.

The slit clamp, part `12-1-bride-fente`, is fitted with two M3 × 10 mm screws. Do not screw the clamp fully home, so as to leave a gap for sliding in the slit; see Figure 24.

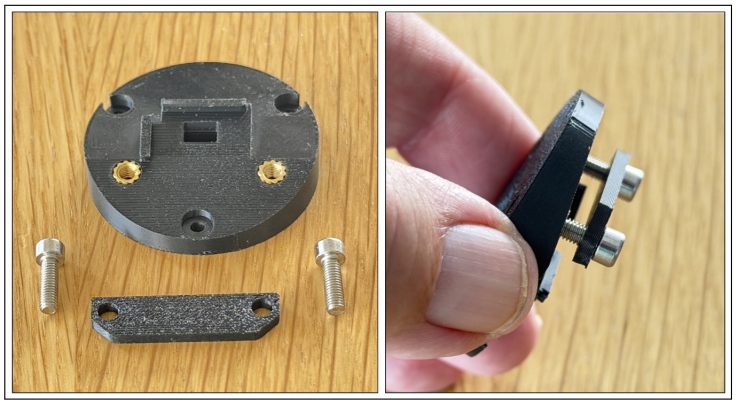


Figure 24. Fitting the slit clamp.

The slit to be used is the four-position GEN2 model (Shelyak Instruments), a glass plate 2 mm thick and 16 mm square (Figure 25).

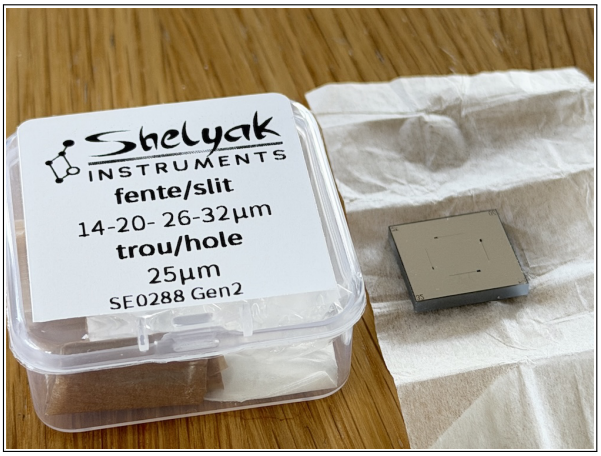


Figure 25. The four-position GEN2 slit (widths of 14, 20, 26 and 32 µm, selected by rotating it through 90°).

The slit can be cleaned. If you spot dust or fingerprints, using a strong magnifying glass or the macro function of your smartphone, do not hesitate to clean it with a cotton bud moistened with a little household alcohol or, better still, isopropyl alcohol (Figure 26). Finish by wiping it gently dry, with two or three strokes in the same direction. Check both faces of the slit.

Figure 26. Cleaning the slit.

Place the slit in the corner formed by the two mechanical stops of the holder (Figure 27).

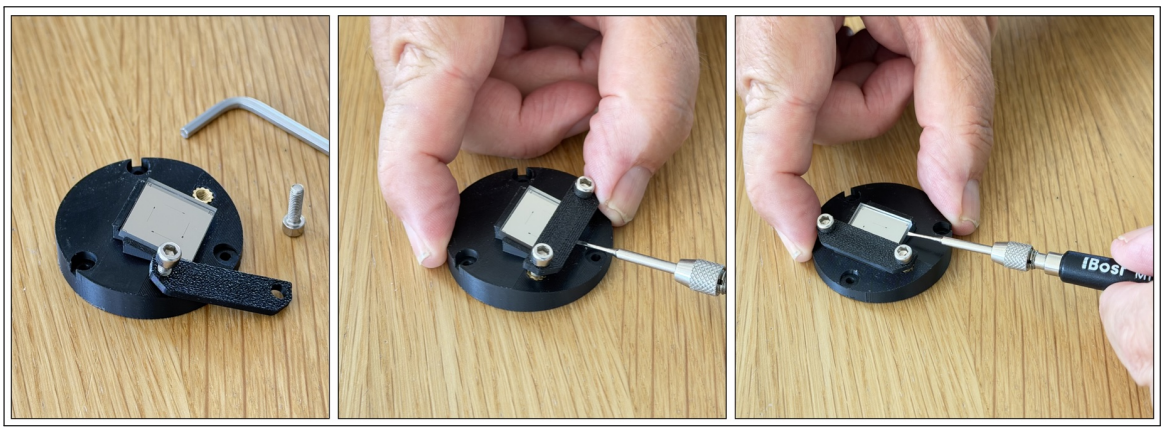


Figure 27. To make it easier to position the slit, you can unscrew one of the two clamp screws. Using a small screwdriver, adjust the position of the slit along both axes of the plane: it must be able to slide, so do not tighten the clamp at this stage. Make sure the slit is pressed firmly against the two walls at 90° that locate it. Look at the slit from the side to check.

The slits are etched in a thin reflective chromium layer deposited on one face of the glass plate. It is **important** that this chromium-coated face faces the holder. The markings on the slit should then read the right way round (Figure 28).

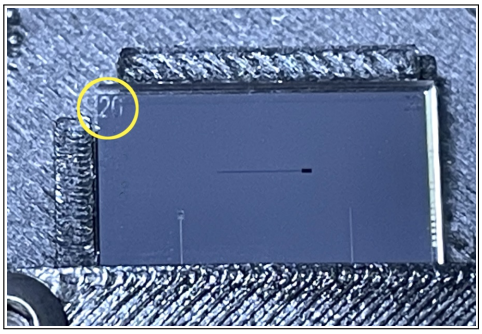


Figure 28. To check that the slit is fitted the right way round, make sure that the markings indicating the widths read correctly. In this example, the 20 µm slit is in position.

Tighten the clamp without excess. It will bend slightly, a sign that it is holding well; but since the grip relies on the flexing of the clamp, the tightening must remain very moderate. Share the effort evenly between the two screws. Then check with the strong magnifier, as described above, that the edges of the slit are in contact with the side stops along their entire length (Figure 29). When starting out with `TinyStar'Ex`, it is recommended to select the 20 µm slit.

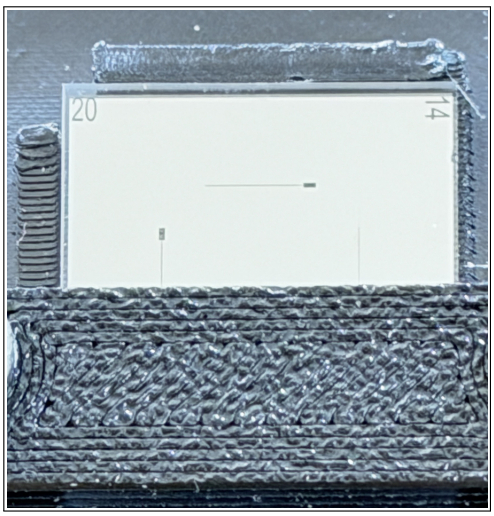


Figure 29. High magnification of the slit after tightening the clamp. The edges of the slit rest firmly against the mechanical stops along their entire length. With the orientation of this example, the 20 µm slit is selected. The photometric zone, here 200 µm wide, can just be made out at one end.

Fix the slit holder to the end of the collimator tube with three M2 screws 8 mm long (Figure 30).

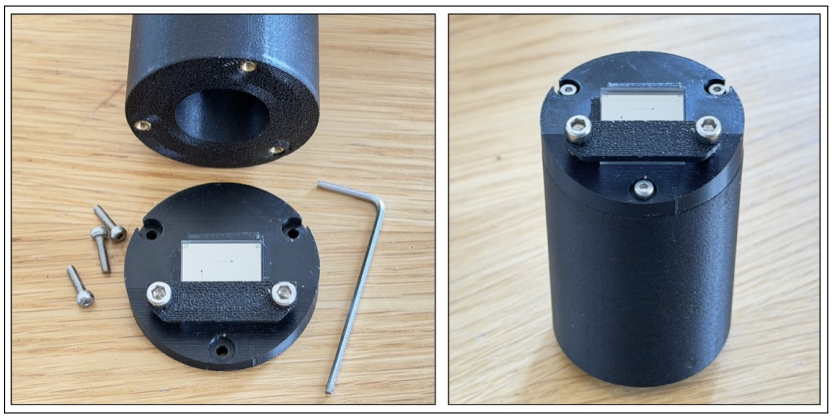


Figure 30. Fitting the slit holder to the collimator tube with three M2 × 8 mm screws. The collimator assembly is now complete.

9.5 The grating holder

The TSE spectrograph nominally uses a ruled grating of 300 lines/mm, 12.7 mm by 12.7 mm and 6 mm thick.

The holder for this grating is part `7-1-porte-reseau-127`. Note, however, that `TinyStar'Ex` can also perfectly well accept the 25 mm by 25 mm, 300 lines/mm grating of the `Star'Ex LR` instrument, by using the `8-1-porte-reseau-250` holder.

The photographs in Figure 31 show the various inserts that complete the grating holder.

Figure 31. The grating holder requires several types of insert. Top: an M3 insert; top right: a (short) M4 S insert; bottom left: an M4 insert; bottom right: three M3 inserts (the position of one of them is hidden by the fingers).

The views in Figure 32 show the grating holder fitted with these inserts.

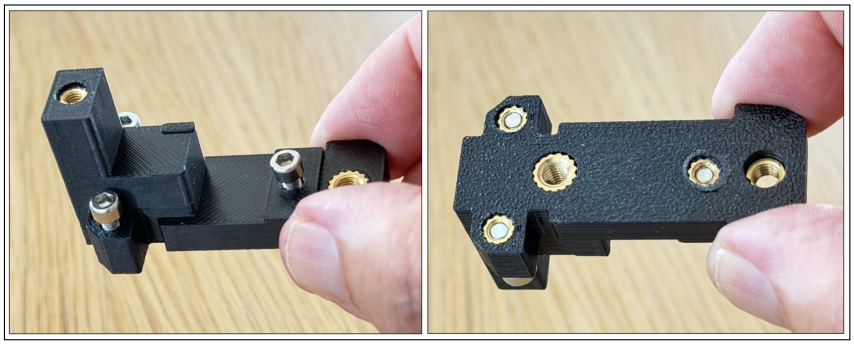


Figure 32. The grating holder ready to receive the grating. The three M3 × 10 mm screws visible in the left-hand photograph are used to correct any tilt of the grating plane.

9.6 Fitting the grating

The grating is the most fragile of all the optical parts. While the surfaces of the lenses or the slit can be wiped and cleaned, this is forbidden for the ruled face of the grating: the very fine grooves on it are damaged by the slightest rubbing. A golden rule: it is better to leave the surface of a grating dirty than to try to clean it. Despite these warnings, do not worry unduly: the assembly is simple and will go well if you are careful and aware of the precautions to take.

The photographs in Figure 33 show how to unpack the grating.

Figure 33. Top left: the grating in its box. Top right: a cut with a craft knife through the adhesive tape opens the packaging. Bottom left: the grating appears; the ruled face is opposite the one you see at this stage. This visible face is flat and has no optical function: you can touch it, and even hold the grating in the palm of your hand by this face. Bottom right: turning the box over onto a sheet of cotton wool or lens tissue, you finally see the ruled face, recognisable by its shiny appearance. This is where you must be careful: no contact whatsoever with this surface.

Always pick up the grating by its edges, then place it in the seat provided for it in its holder (Figure 34).

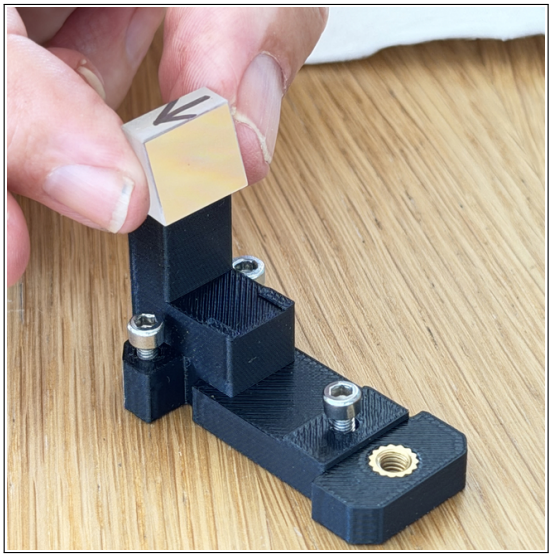


Figure 34. Handling the grating: always by its edges. You may wear gloves, but even with gloves, never touch the ruled face, the one that looks shiny to the eye.

In the left-hand photograph of Figure 35, the grating is in place. Note carefully that the ruled face faces outwards. Note also the orientation of the arrow drawn on the edge of the grating relative

to the holder: it is **essential** to respect this orientation, as the efficiency of your spectrograph depends on it (it allows you to benefit from the *blaze* of the grating)[3].

The grating is held by the clamp `9-1-bride-reseau`, fixed with an M3 × 8 mm screw (right-hand photograph of Figure 35). This clamp has a small shoulder that holds the grating more securely: do not fit it the wrong way round. Tighten without excess, then check, holding the grating by its edges, that it no longer moves and is properly centred. Also check that its back rests along its entire length against the upright of the holder. This is important, as the centring of the spectrum on the sensor that records it depends on it.

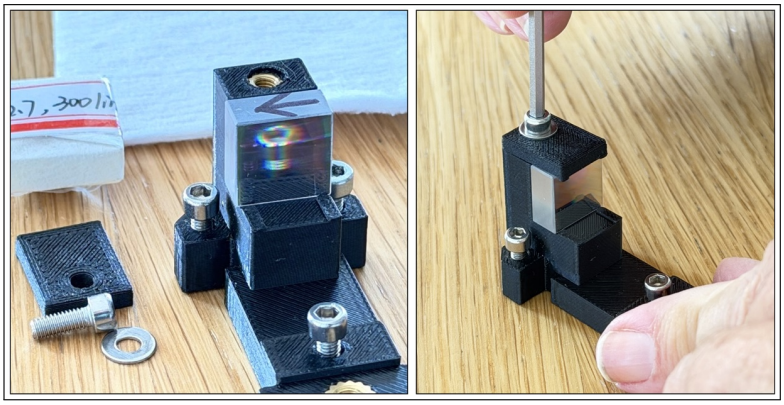


Figure 35. The grating positioned in its holder. A thin lip in the holder helps to keep the grating pressed against its rear face and secure in position. Note the direction of the arrow carefully (it is the orientation that matters, not the edge on which it is drawn). The grating clamp is in place (M3 × 8 mm screw). A small lip on part `9-1-bride-reseau` also makes it easier to position the grating and holds it more firmly.

Check once more that the rear face of the grating is pressed firmly against the upright (Figure 36). If it is not, loosen the clamp and adjust the contact.

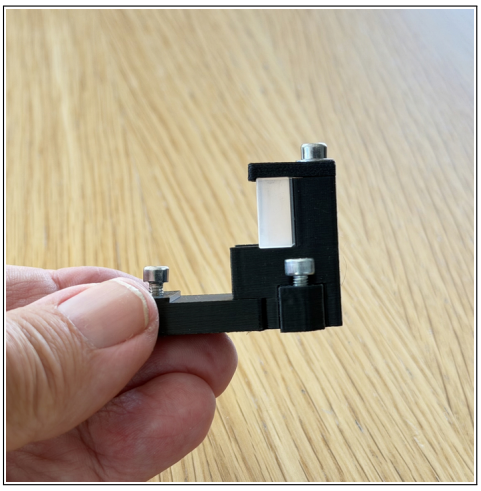


Figure 36. Checking that the grating is properly seated.

Fit part `10-1-interface-reseau` with two M4 inserts and two M4 × 20 mm screws, which will provide the fine orientation of the grating (Figure 37).

[3] The arrow may sometimes be drawn on the opposite edge. It is then at the bottom, hidden by the grating holder. What matters, however, is still the orientation of the arrow, as shown in Figure 35.

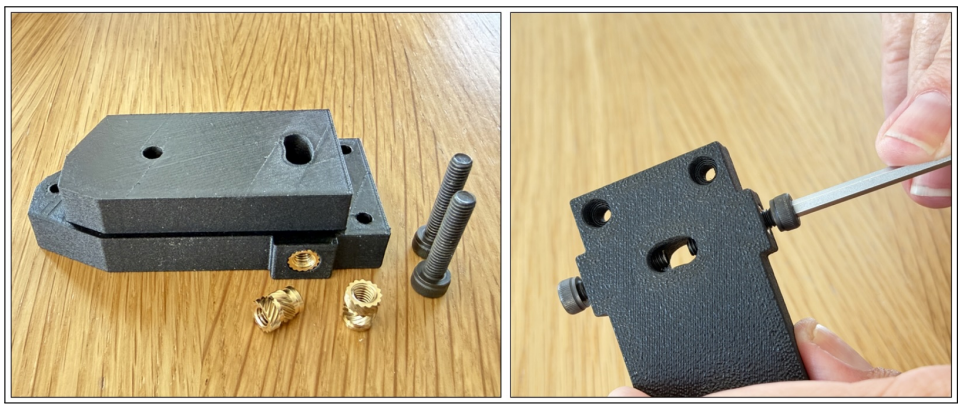


Figure 37. Completing part `10-1-interface-reseau`: two M4 inserts to be fitted and a pair of M4 screws 20 mm long.

Add an M4 screw 30 mm long to part `7-1-porte-reseau-127`, as shown in Figure 38. Do this only after fitting the grating: fixing the grating is much easier without the protrusion formed by this long screw.

Reduce the risk of contact with the ruled surface of the grating by setting this assembly down as shown on the right of Figure 38.

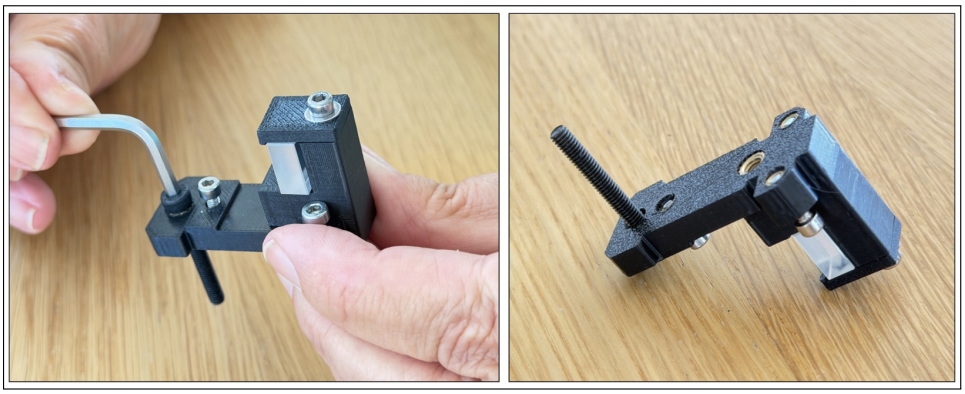


Figure 38. Left: adding an M4 screw 30 mm long to the grating holder. Tighten it firmly, as this screw acts as a stud for the fine adjustment of the grating. Always keep your movements measured, and never touch the front face of the grating. Right: how to set down the assembled grating holder, ruled face downwards for safety.

Attach the grating holder to its interface as shown in Figure 39. Then put this assembly aside, protected from dust and accidental contact.

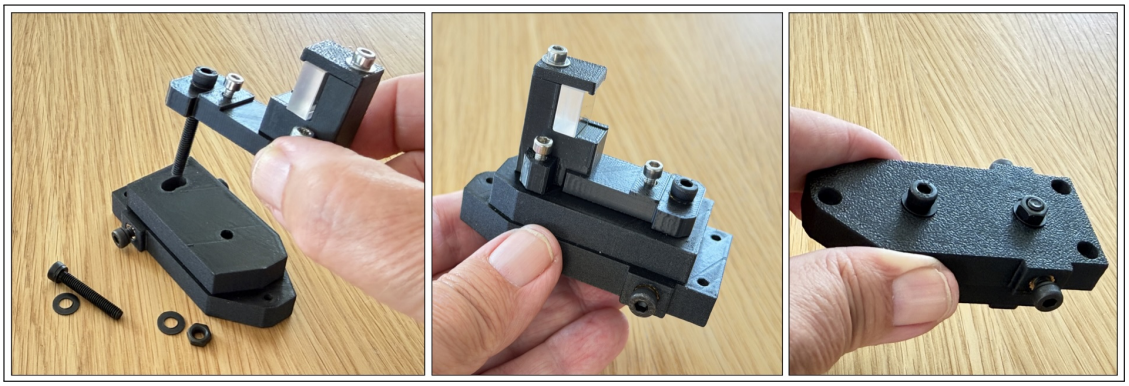


Figure 39. How to mount the grating holder on its interface. You will need an M4 screw 25 mm long, which acts as a pivot when the grating is rotated (the axis of this screw lies exactly in the plane of the ruled surface). You will also need an M4 nut (or, optionally, an M4 Nyloc nut, which will not work loose). Do not forget to use washers, to make adjustment easier and avoid damaging the plastic when tightening (always moderately).

9.7 Closing the housing

Screw the camera tube assembly firmly onto the camera itself via the T2 interface. Adjust the distance between the front of the camera tube and the mounting face of the camera (here a Player One Uranus-M Pro) as shown in Figure 40.

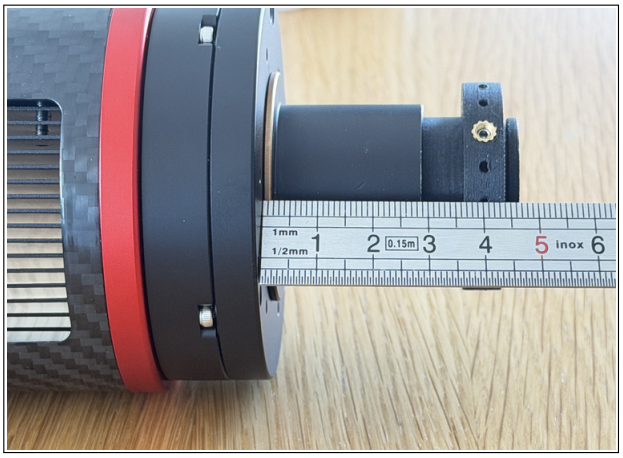


Figure 40. By screwing the camera tube in or out, adjust the distance between the mounting face of the camera (itself 17.5 mm from the detector, a standard dimension) and the front of the tube to about 44 mm.

Place the collimator and camera tubes in their seats in the lower part of the housing (see Figure 41).

Figure 41. Positioning the components before closing the housing. You will need five M4 screws 60 mm long, with their washers.

Fix the two shells of the housing together. Do not tighten the screws firmly at first: the collimator and camera tubes must be able to turn freely in their seats, with only slight resistance.

As highlighted in Figures 42 and 43, make sure that the 31.75 mm barrel carrying the camera and the collimator tube are in contact with their longitudinal locating shoulders.

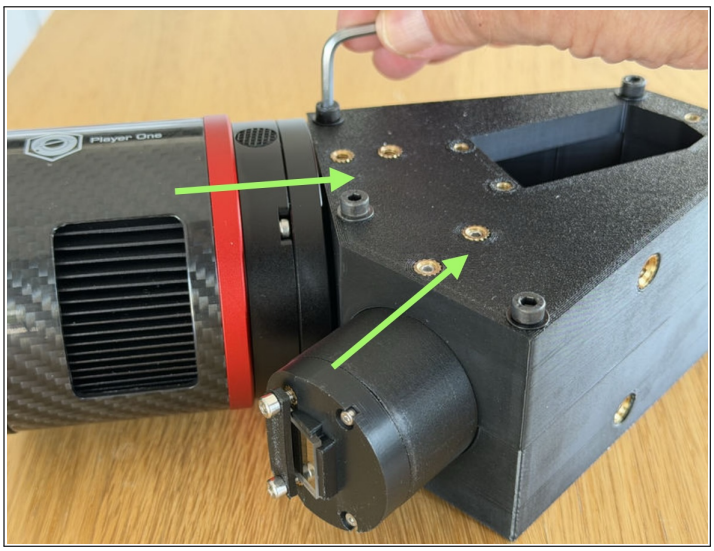


Figure 42. Assembling the two shells of the housing. It is **important** to push the components firmly in the direction of the green arrows, so that they bear against their shoulders and are therefore in their nominal positions. Note also the orientation of the collimator tube, using the slit holder as a reference: **the inclined face points inwards**.

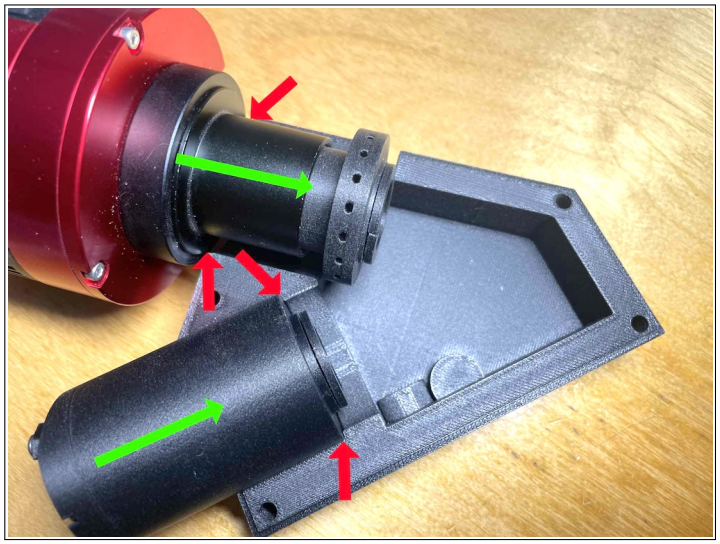


Figure 43. It is important that the sliding components sit properly on their mechanical bearing surfaces after the two shells have been assembled and tightened, including after a rotational adjustment (red arrows).

Next, fit the grating assembly in its seat, which almost closes the TSE housing (Figure 44).

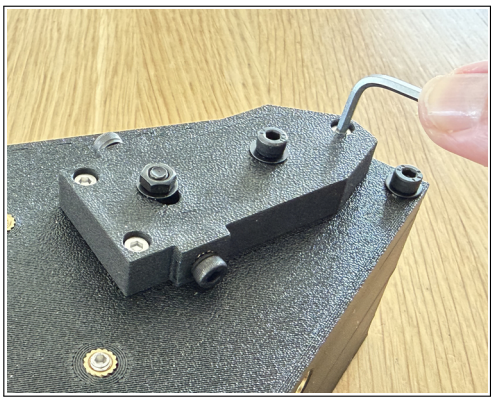


Figure 44. Fitting the grating interface, using three M3 screws 12 mm long.

10 Adjusting the spectrograph

10.1 The method

You can now record your first spectrum with `TinyStar'Ex`. Daylight is all you need; clear weather gives more light, but this is not essential.

All the adjustments of the spectrograph are made using the solar spectrum. This has two great advantages: it is a natural source, available at any time of day, and its spectrum is crossed by numerous dark lines (the Fraunhofer lines) whose wavelengths are precisely known. These lines serve as references for focusing and for checking that the instrument is working correctly.

Everything is done without mounting the spectrograph on the telescope: it is simply placed on a table, and you aim at the sky through a window.

Do not point the spectrograph directly at the Sun: too bright and too small, it would illuminate the slit very unevenly, which is unsuitable for the adjustments. Aim instead at the blue sky, a bright cloud or a well-lit wall: the scattered light contains the solar spectrum we need.

Remember the essential point: most of the adjustments are made in daylight, without a telescope, calmly and in comfort. Once the instrument has been properly adjusted in this way, obtaining fine stellar spectra at night becomes much simpler, almost a formality.

10.2 Orienting the camera

Connect the camera to the computer with a USB cable and start your usual acquisition software. Take an exposure of a few tenths of a second, without saturating the image, with a gain of 200 (20 dB) and an offset of 20 (typical values for an Uranus camera). At first the spectrum is blurred and wrongly oriented: this is normal, and we shall put it right. Carry out the adjustments in the order given below.

Begin by adjusting the orientation of the camera. Figure 45 shows what happens when this orientation is incorrect, and how to remedy it. Slightly loosen the screws joining the two shells of the housing, so that the camera can be turned in the 31.75 mm barrel with slight friction. Check the result live on the computer screen. With practice, you will be able to judge a tenth of a degree; but at this stage, an approximate adjustment is enough.

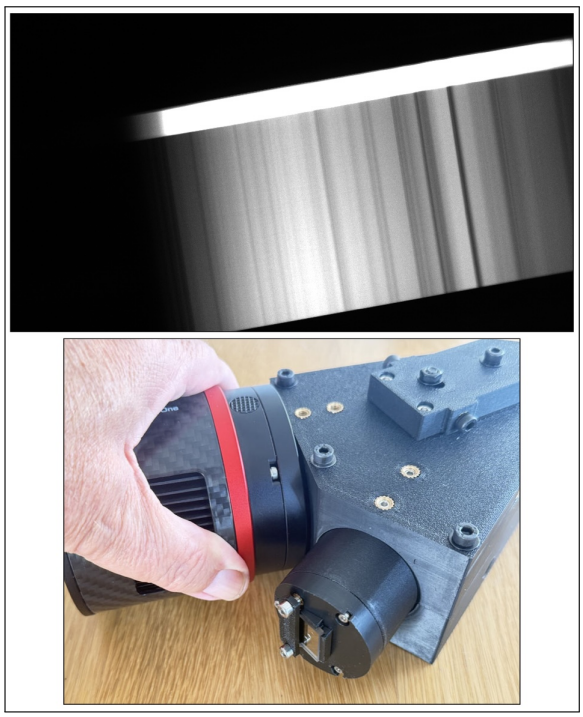


Figure 45. Orienting the camera to make the dispersion axis horizontal (also known as the *tilt* adjustment). In this view, the axis is steeply inclined: it is the line rising from left to right, which must be made parallel to the pixel rows of the sensor. Take advantage of this first image to check that the blue end of the spectrum is on the left and the red end on the right. This is **mandatory**. If it is not the case, change the camera's readout settings or rotate it through 180°.

You have now approximately aligned the dispersion axis of the spectrum with the pixel rows of the sensor.

10.3 Focusing the spectrum

At this stage, the lines of the solar spectrum are probably blurred, or even barely visible: the image of the spectrum is not yet correctly focused on the detector. To focus, turn the camera lens tube by trial and error, first by half a turn, then by a quarter of a turn, and finally by two or three tenths of a turn (Figure 46). Depending on the direction of rotation, the lens moves forwards or backwards. You reach the adjusting ring through the hatch provided for the purpose: insert a fine screwdriver or an Allen key in the holes of the ring to turn the camera tube. You will no doubt turn it the wrong way at first, but you will end up with a sharp spectrum.

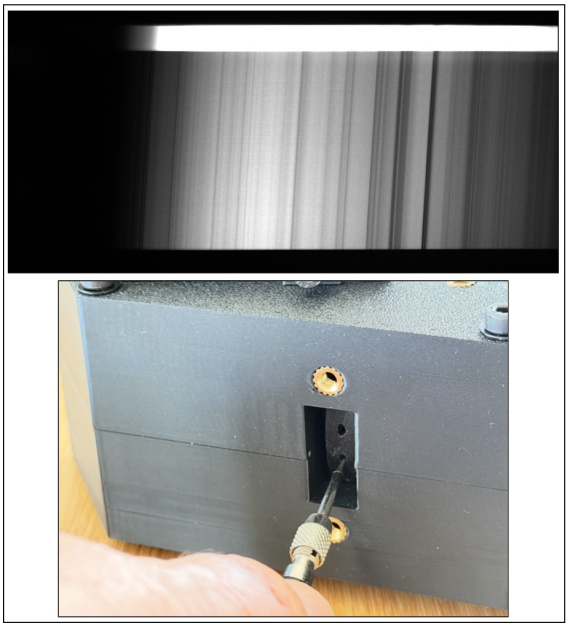


Figure 46. Adjusting the sharpness of the spectrum.

Once the spectrum is sharp, its upper and lower edges are clearly visible. Take this opportunity to refine the orientation of the camera so that these edges are as horizontal as possible. An error of 1° is too large: aim for an accuracy of a few tenths of a degree, which can be judged by eye by examining the spectrum along its whole length. Then lock the rotation of the camera by tightening, without excess, the two Teflon-tipped thumbscrews of the 31.75 mm barrel. The residual angle between the dispersion axis and the pixel rows of the sensor is called the 'tilt angle'.

10.4 Making the lines vertical

After adjustment, the spectral lines must be truly vertical, that is, perpendicular to the dispersion axis. The aim here is to correct their inclination relative to the columns of the detector, known as *slant* (Figure 47). This adjustment is made by rotating the collimator tube, after loosening the housing screws on the collimator side.

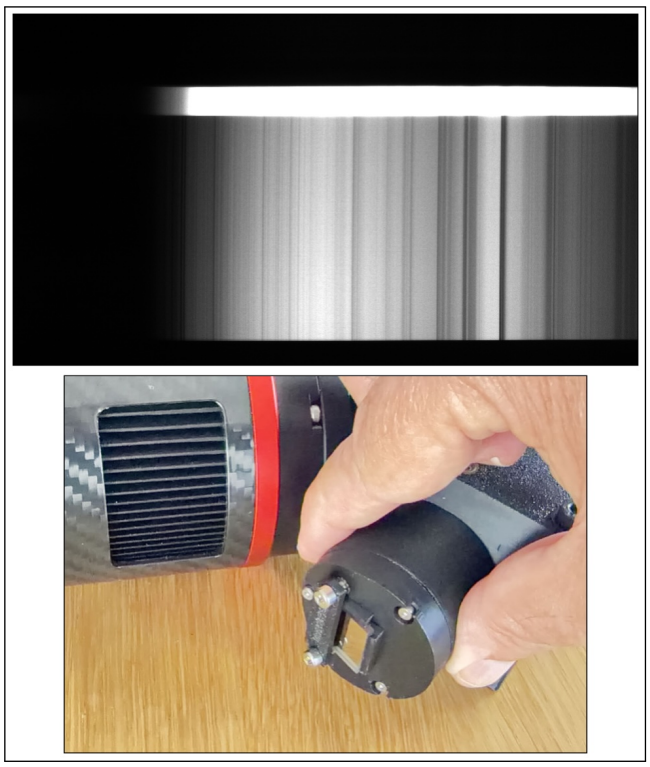


Figure 47. Adjusting the inclination of the spectral lines.

This adjustment is made by hand and calls for a light touch. Start a continuous acquisition loop and follow the result live on the spectrum image displayed on screen. Adjust the tightness of the screws so that the part turns with slight friction: it is then easier to handle and holds its position. Proceed by trial and error, and be patient. Normally, you will not need to repeat this adjustment.

If you cannot achieve satisfactory verticality by hand, use the small tools provided, the parts `1A-fourche-reglage` and `2A-manette-reglage` (adjustment fork et adjustment lever), as shown in figure 48.

The corresponding print files are gathered in a compressed archive, `stl_outils.zip`. The associated OpenSCAD file, `TSE_sources_outils.scad`, can be downloaded in the same way.

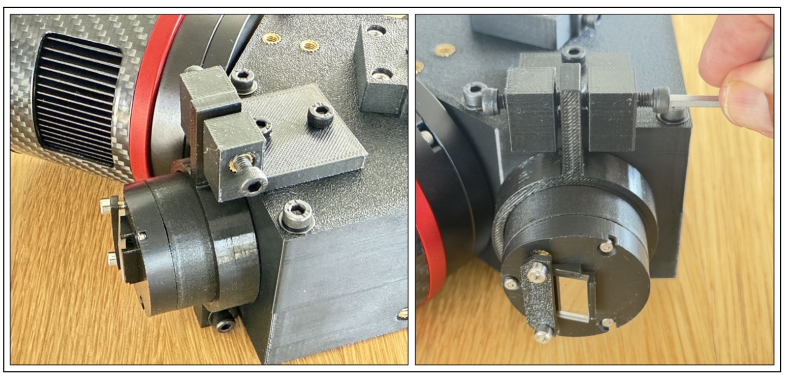


Figure 48. Tools to help rotate the collimator tube.

To work comfortably afterwards, both on the sky and during processing, aim for a *slant* error of less than a few tenths of a degree. An error of 1° is clearly visible if you take a good vertical reference in the image, and must be corrected.

10.5 Positioning the spectrum horizontally

The instrument only gives truly sharp lines if the spectrum is placed in the right position along the dispersion axis. First locate the Hα line (Figure 49).

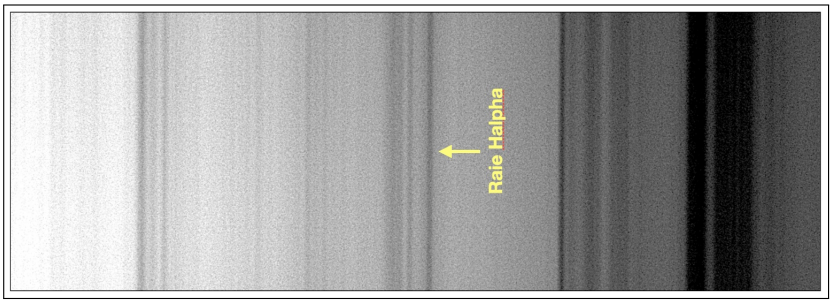


Figure 49. Locating the Hα line in this part of the solar spectrum.

With a camera fitted with an IMX585 sensor, this line should lie 2450 pixels from the left-hand edge of the image, to within $\pm$50 pixels. It is normal for the spectrum then to appear shifted to the right: this is the position in which you will obtain the best results. To move the red hydrogen line horizontally, and with it the whole spectrum, rotate the grating about an axis lying in its plane, as shown in Figure 50. Once the position has been reached, retighten the two M4 screws moderately.

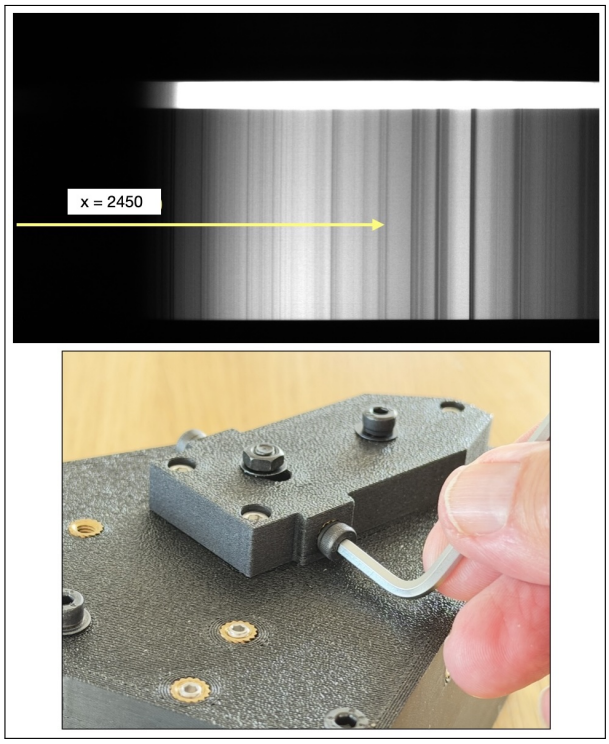


Figure 50. Fine rotation of the grating to position the spectrum on the sensor. First slightly loosen the M4 pivot screw and the nut of the adjusting rod (the screw that passes right through the instrument). Then use the two screws that push the arm of the grating holder: they work in opposition. The case shown in this figure corresponds to the use of a Sony IMX585 sensor.

Suppose the sensor of your camera has L pixels along its longer dimension, and that its pixels measure p μm. The Hα line of the solar spectrum (daylight) should then be placed at the position:

$$x = \frac{L}{2} + \frac{1514}{p} \quad \text{(in pixels).}$$

Take the example of a camera fitted with a Sony IMX533 sensor: L = 3008 pixels and p = 3.76 µm, giving x = 3008/2 + 1514/3.76 ≈ 1907 pixels, to within ±50 pixels.

10.6 Centring the spectrum vertically

If the spectrum is not sufficiently centred vertically in the image, use the levelling screws of the grating holder (Figure 51). Proceed by trial and error to find the direction and extent of the adjustment.

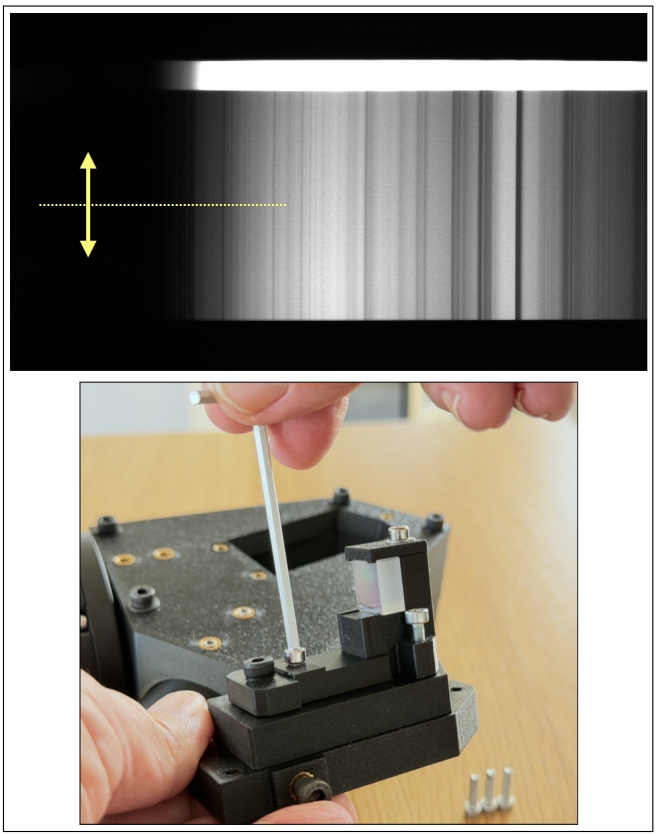


Figure 51. Adjusting the vertical position of the spectrum. Do not force the part: first loosen its fixing screws on the interface to give a little play.

10.7 Finishing touches

Tighten all the screws joining the two shells of the housing, and also, moderately, the locking screws of the collimator and camera tubes (Figure 52).

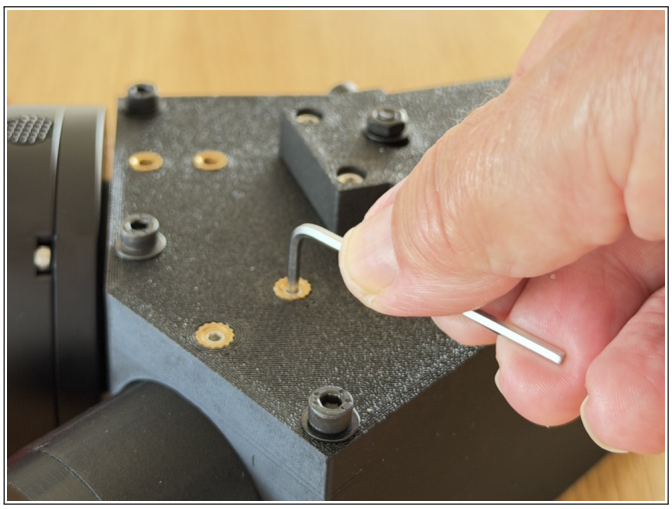


Figure 52. Final tightening of the Teflon-tipped grub screws.

Finish closing the housing by fitting part `13-1-trappe-focus` (Figure 53). It opens easily if, when observing stars, the focus of the spectrum proves inadequate. This is the only adjustment of the spectrograph that may need to be repeated on the telescope, and it is a simple one.

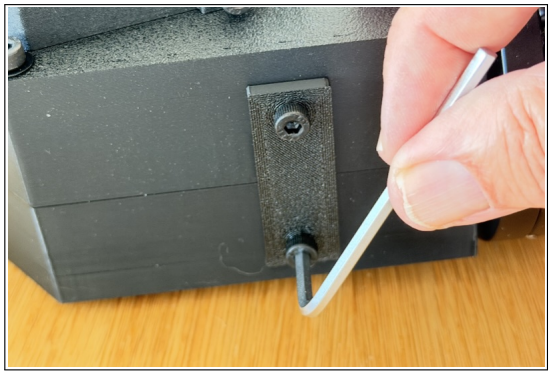


Figure 53. Closing the access to the focus adjustment (two M4 × 8 mm screws).

You have now completed the construction and assembly of the spectrograph itself. One last step remains before you can capture the spectra of celestial objects: building and fitting the system for pointing at the slit.

In the end, the spectrograph will look like the one shown in Figure 54.

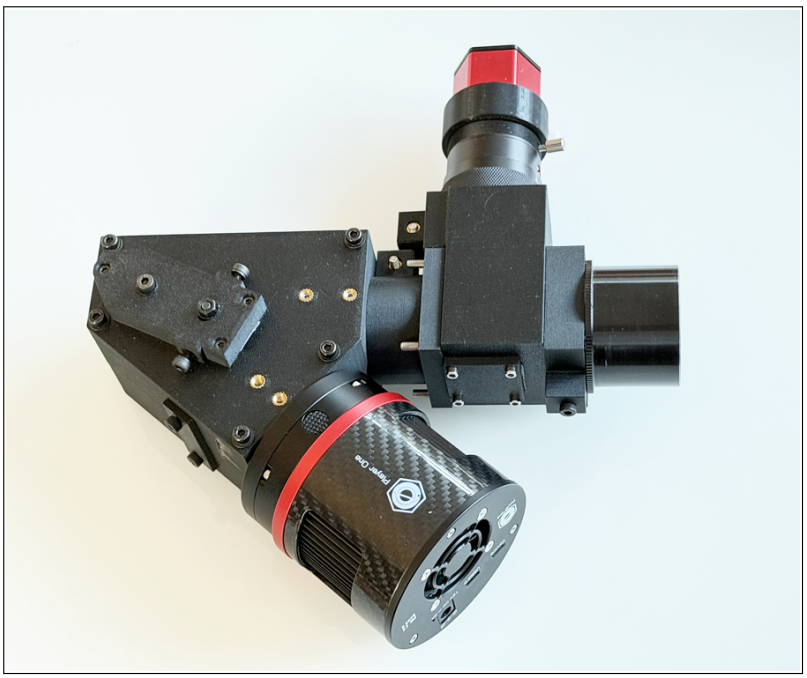


Figure 54. The complete TinyStar'Ex spectrograph, fitted with its pointing and guiding system.